MILITARY SPECIFICATION

MOUNT, PERISCOPE: M119

1. SCOPE

1.1 This specification covers one type of instrument known as Mount, Periscope, M119 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8619585	Mount, Periscope: Ml19
F11738158	Gage, Interchangeability, Maximum
F11738159	Gage, Interchangeability, Minimum

PACKAGING DATA SHEET

8619585	Packaging of Mount, Periscope: M119

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(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The mount shall be manufacted in accordance with Drawing F8619585 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following erquirements of MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.2.1 Interchangeability.- The mount shall be capable of permitting insertion of the gages specified in 4.6.7. After insertion, the gages shall be capable of being firmly secured and subsequently removed.

3.3 Environmental.

3.3.1 Operating temperatures.- The mount shall meet the applicable requirements of 3.5.1 to 3.5.4.3 inclusive while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.3.2 Vibration.- The mounst shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 ± 1/4 minutes. Subsequent to vibration, there shall be no evidence of physical failure and the mount shall meet the following requirements of this specification (see 3.5.4.3).

3.4 Orientation.- The locating surface of the mounting flange of the mount shall be secured in a horizontal plane and the headrest assemblies shall be in their uppermost vertical position. Gages referenced in 3.2.1 shall be utilized to determine compliance with the requirements of 3.5.1 and 3.5.2.

3.5 Performance.

3.5.1 Safety latch.- With the gages in 3.2.1 inserted into the mount, the safety latch shall snap under the wing nut screw so that the gages cannot be removed unless the safety latch is actuated. Actuation of the safety latch shall allow removal of the gages from the mount without restriction-

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3.5.2 Holding features. - When both wing nuts are screwed down until tight against the mount surface, the gages (see 3.2.1) shall be secured in the mount. Subsequent to the application of a 10 pound pressure applied horizontally to the front and rear of the gage at a point 7.5 inches below the center of the wing nut screws, the gage whall be within .0005 inch of its original location when measured horizontally at a point 7.5 inches below the center of the wing nut screws. This requirement shall be met following the release of each of the pressures applied.

3.5.3 Shield assembly.- With the handle lock disengaged (turned toward the gage), maximum vertical movement of the handle shall operate the shield through the full travel range (closed for protection and open for viewing). When the shield is fully closed, the handle shall engage and lock. The handle shall also engage the intermediate stops through the full travel range.

3.5.4 Headrest.

3.5.4.1 Headrest, horizontal and rotational travel.- When the locking levers are in the unlocked position, both headrests shall be adjustable along and about the support shafts. When the locking levers are turned to the locked position, both headrests shall be firmly secured to the support shafts.

3.5.4.2 Headrest, vertical travel.- With Knob 10556521 loosened, the headrest (see Drawing D8619420) shall be adjustable through maximum movement permitted by the elongated slot in the assembly without restriction or force.

3.5.4.3 Retention. - The torsional load applied to each Screw 8619691 shall be such that the headrests shall not change position when subjected to the vibration specified in 3.3.2 and shall be operable at the temperatures specified in 3.3.1.

4. QUALITY ASSUraNCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 mounts plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicalbe Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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4.3.2.2 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Interchangeability	3.2.1	4.6.7
Safety latch	3.5.1	4.6.3

TABLE I - CLASSIFICATION OF DEFECTS

CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Vibration	3.3.2	4.6.1.2
102. Holding features	3.5.2	4.6.4
103. Shield assembly	3.5.3	4.6.5
104. Headrest, horizontal and rotational travel	3.5.4.1	4.6.6.1
105. Headrest, vertical travel	3.5.4.2	4.6.6.2
106. Retention	3.5.4.3	4.6.6.3

MINOR: NONE

NOTE: The tests of Table I shall be conducted at a temperature between + 60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One mount shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawing - visual
302. General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III.

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TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Safety latch	3.3.1, 3.5.1	4.6.1.1, 4.6.3
304.	Holding features	3.3.1, 3.5.2	4.6.1.1, 4.6.4
305.	Shield assembly	3.3.1, 3.5.3	4.6.1.1, 4.6.5
306.	Headrest horizontal and rotational travel	3.3.1, 3.5.4.1	4.6.1.1, 4.6.6
307.	Headrest vertical travel	3.3.1, 3.5.4.2	4.6.1.1, 4.6.6.2
308.	Retention	3.3.k, 3.5.4.3	4.6.1.1, 4.6.6.3

NOTE: The tests for Characteristics 303 through 308 inclusive shall be conducted at -40° and +150°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4*3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming-a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed

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90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The mount shall be exposed to the temperature extremes specified in 3.3.1 for a minimum of 2 hours at each extreme. At the end of each 2 hour period and while still at each temperature extreme, the mount shall be subjected to the tests specified in 4.6.3 to 4.6.6.3 inclusive.

4.6.1.2 Vibration.- The mount shall be secured to a medium and vibrated as required by 3.3.2. Subsequent to the vibration a visual and tactile inspection shall be made for physical damage and loose components or assemblies. After which the mount shall meet all the requirements specified herein.

4.6.2 Orientation. - A testing device (see Figure I), shall be fabricated to support and position the mount as specified in 3.4, with enought area to allow the insertion of and removal of the gages specified in 4.6.7. The testing device shall also provide a means of applying and measuring pressure from opposite directions as required by 3.5.2.

4.6.3 Safety latch. - The gages specified in 4.6.7 shall be utilized for this test. The mount shall be positioned in the testing device (Figure I) with the headrest in the upper most position and secured. The maximum interchangeability gage shall be inserted upward into the mount until the safety latch is activated and snaps into position. Allow the safety latch to hold the gage in the mount. (caution should be exercised before releasing or removing hands from medium to prevent injury, if safety latch fails.) To remove the gage, the inspector will exert an upward pressure with one hand and activate the spring loaded safety latch with the other. The gage can be lowered sufficiently with the one hand and then removed from the mount as specified in 3.5.1. Perform the above test using the minimum interchangeability gage to determine conformance with 3.5.1.

4.6.4 Holding features. - Each gage shall again be inserted into the mount as in 4.6.3. A dial indicator (Figure I) graduated in .0001 increments shall be positioned so that it contacts at either the front or rear of the gage at a point specified in 3.5.2 and adjusted to read zero with the indicating hand. Apply a slow steady pressure of 10 pounds to the opposite side from the dial indicator and then upon releasing the pressure, the indicating hand of the dial indicator shall not exceed the tolerance specified in 3.5.2. Reverse the equipment and

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make the second reading by reversing the above procedure.

4.6.5 Shield assembly.- With the mount positioned as i 4.6.3, the inspector shall rotate the pistol grip of the assembly in a clockwise direction and push up on the grip to open the protecting shield to the maximum position. Release the grip will activate the torsion spring that will lock the assembly in the notch on the shaft. The three positions between full open and close will be checked for ease of operation as required by 3.5.3.

4.6.6 Headrest.

4.6.6.1 Headrest, horizontal and rotational travel.- This operability check can be conducted with the set-up specified in 4.6.5. Each headrest shall travel the full length of the shaft horizontally, when locking lever is loosened, and shall rotate freely around the shaft to insure available adjustment at any position. When the locking levers are in the locked position the headrest shall be secured at any position on the shaft as required by 3.5.4.1.

4.6.6.2 Headrest vertical travel.- This operability check can be conducted in conjunction with 4.6.6.3. The assemblies shall be adjustable without interference or forcing as required by 3.5.4.2.

4.6.6.3 Retention. - The headrest assemblies shall be positioned at a known point and secured prior to the vibration test. After the vibration is complete the headrest shall not have moved or changed position. as required by 3.5.4.3.

4.6.7 Interchangeability. - This test shall be performed utilizing the maximum and minimum gages conforming to Drawings F11738158 and F11738159 respectively, to determine compliance with 3.2.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8619585. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Periscope Mount, M119 is used to support either the Periscope, Tank: M34 or periscopes Tank: M36.

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6.2 Orderind data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (See 5.1).

Custodian:

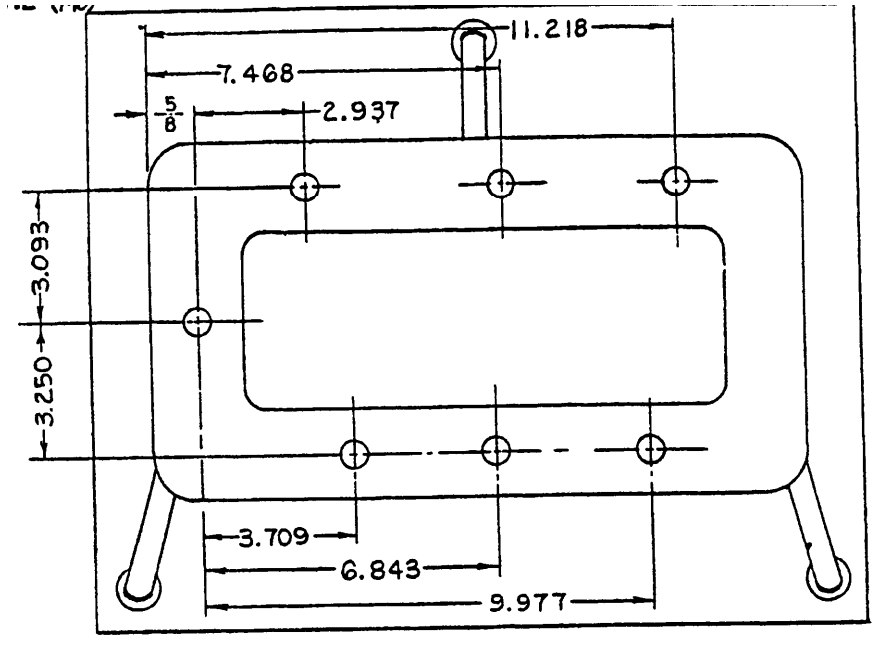
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Preparing activity:

Army - MU

Project No. 1240-A263

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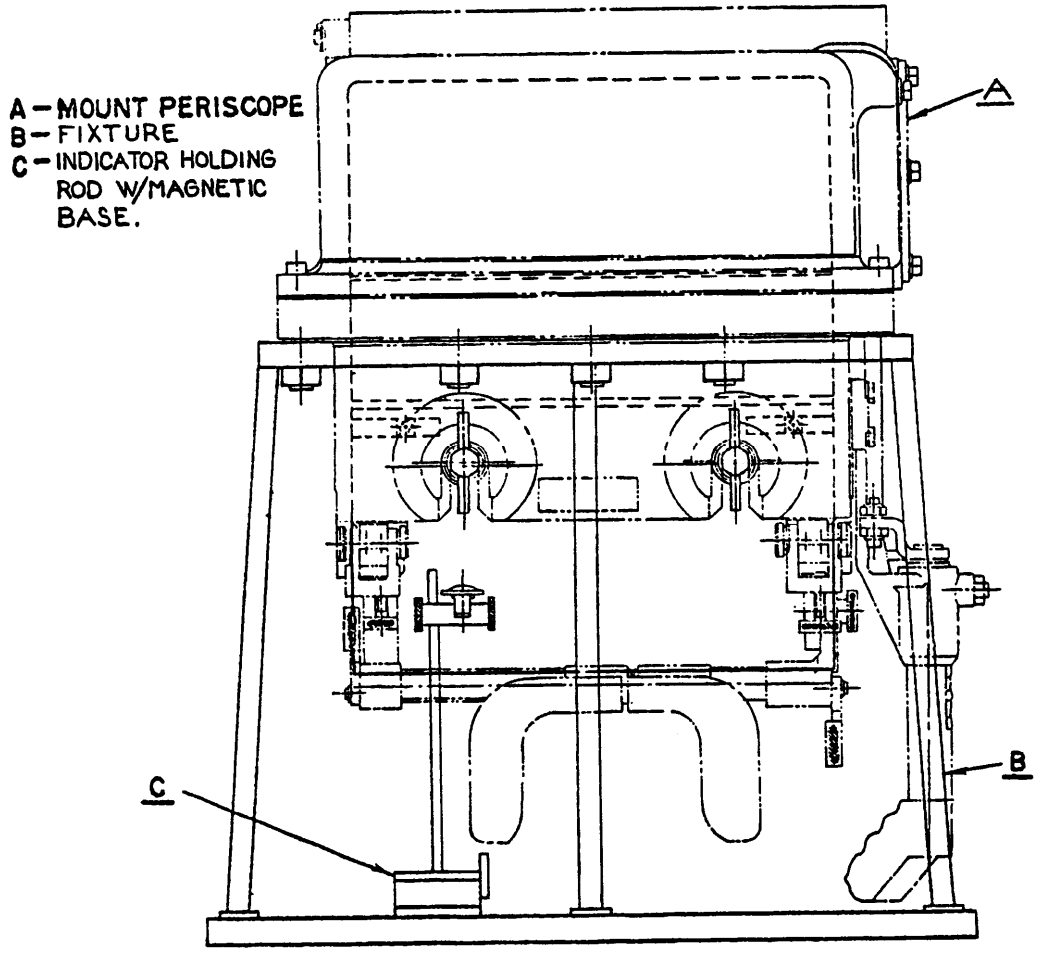


FIGURE I

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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